Solution

1 The problem

Let $D \subseteq \mathbb{R}^d$, $d \geq 2$, be the closed Euclidean unit ball. For $f \in C^\alpha(D)$ with Hölder–Zygmund smoothness $\alpha > 1$ (and f bounded away from 0), let $u = u_f$ be the unique solution of Darcy's problem

$$\operatorname{div}(f\nabla u) = 1 \ \text{ on } D, \qquad u|_{\partial D} = 0.$$

We observe

$$Y_n = u_f + \tfrac{1}{\sqrt{n}}\,\mathbb{W},$$

where $\mathbb{W}$ is a Gaussian white noise on $L^2(D)$. For $r \in (0,1)$ let

$$\mathcal{F}(\alpha, r) = \{h \in C^\alpha(D) : \ \|h - \mathbf{1}\|_{C^\alpha(D)} \leq r\},$$

where $\mathbf{1}$ is the constant function 1. For small but fixed $r \in (0,1)$, the question is: what is the minimax rate, as $n \to \infty$ and up to logarithmic factors, of

$$\inf_{\hat{f}_n} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0}\big[\|\hat{f}_n - f_0\|^2_{L^2(D)}\big] \ ?$$

2 Result

Theorem 2.1. *Fix $\alpha > 1$ and $d \geq 2$.*

(i) ***(Lower bound.)*** *For every $r \in (0,1)$ there are $c > 0$ and n_0 such that for $n \geq n_0$*

$$\inf_{\hat{f}} \sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \|\hat{f} - f\|^2_{L^2(D)} \ \geq \ c\, n^{-2\alpha/(2\alpha+d+3)}.$$

(ii) ***(Upper bound.)*** *There is $r_0 = r_0(d,\alpha) > 0$ such that for every $r \in (0, r_0]$ an explicit estimator $\tilde{f}$ (Definition 6.2) satisfies, for $n \geq n_0$,*

$$\sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \|\tilde{f} - f\|^2_{L^2(D)} \ \leq \ C(\log n)^4\, n^{-2\alpha/(2\alpha+d+3)}.$$

If α is not an integer, the factor $(\log n)^4$ can be dropped.

The constants r_0, c, C, n_0 depend only on d, α, r and on fixed auxiliary profiles and kernels chosen below.

Corollary 2.2 (Answer)**.** *For every fixed $r \in (0, r_0]$, up to logarithmic factors,*

$$\inf_{\hat{f}_n} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0} \|\hat{f}_n - f_0\|^2_{L^2(D)} \ \asymp \ n^{-\frac{2\alpha}{2\alpha+d+3}}.$$

Everything in Sections 5–6 is argued here; the only external inputs are the standard facts collected, with references, in Section 4.

3 Overview of the proof

Where $d+3$ comes from. Near $f = \mathbf{1}$ the solution is close to $u_1 := (|x|^2-1)/(2d)$, so $\nabla u_f \approx x/d$, a radial field with a single source at the origin. The linearised forward map $h \mapsto -L_f^{-1}\operatorname{div}(h\nabla u_f)$, with $L_f = \operatorname{div}(f\nabla\cdot)$ (Dirichlet), therefore smooths by order 2 for perturbations oscillating across the rays (in the $d-1$ angular directions), but only by order 1 along the rays, because $\operatorname{div}(h\,x/d)$ is a first-order operator along the rays. Heuristically, the symbol of the linearisation at $f = \mathbf{1}$ is $\asymp (1+|\xi_{\mathrm{rad}}|)/|\xi|^2$, and the exponent $d+3$ lies between the isotropic values $d+4$ and $d+2$. This heuristic only motivates the construction; it is not used in the proofs.

Lower bound. We perturb $\mathbf{1}$ by $M \asymp 2^{j(d-1)}$ disjoint bumps that oscillate at frequency 2^j in the angular directions only. Each has the form $2^{-j(\alpha+2)}b(|x|)\Delta\Phi_k$ with Φ_k homogeneous of degree 0; because of this form, the radial transport term $\operatorname{div}(h\,x/d)$ is again a Laplacian of a function of size $2^{-j(\alpha+2)}$. A duality argument then shows that the forward map shrinks each bump by 2^{-2j} in L^2. Balancing the Kullback–Leibler divergence with Assouad's lemma gives $2^j \asymp n^{1/(2\alpha+d+3)}$ and risk $2^{-2j\alpha}$.

Upper bound. We smooth the data at resolution 2^{-J}, $2^J \asymp n^{1/(2\alpha+d+3)}$, and estimate ∇u by $\hat V = \nabla K_J Y$. For r small, $\hat V$ is C^1-close to x/d, and the transport equation $\operatorname{div}(g\hat V) = 1$ has an explicit solution $T_{\hat V}^{-1}\mathbf{1}$, obtained by integrating along the backward flow of $\hat V$, which runs into a single sink. We linearise around the deterministic centring function $\bar f = T_{V_J}^{-1}\mathbf{1}$, where $V_J = \nabla K_J u$. This gives an exact decomposition: a bias $\bar f - f = O(J^2 2^{-J\alpha})$, a linear Gaussian term Z and a quadratic remainder. The variance of Z is the key point. Z integrates $\Delta\xi$ along rays, where ξ is the smoothed noise. Now $\Delta\xi$ has pointwise variance $n^{-1}2^{J(d+4)}$ and correlation length 2^{-J}, and integrating it along a ray of length $O(1)$ gains one factor 2^{-J}. Hence $\mathbb{E}\|Z\|_2^2 \lesssim n^{-1}2^{J(d+3)}$: the noise is smoothed by one order along the flow, and this produces $d+3$. The quadratic remainder is negligible because $\alpha > 1$.

4 Conventions and standard facts

C^s denotes the Hölder–Zygmund space $B^s_{\infty\infty}$; on D it consists of restrictions, with the quotient norm. Constants c, C may change from line to line. They never depend on j, J, n or on $f \in \mathcal{F}(\alpha, r)$. D° denotes the open unit ball. We write $u_1 := (|x|^2-1)/(2d)$, the solution for $f = \mathbf{1}$, so that $\nabla u_1 = x/d$. For $g \geq 1-r$ write $L_g = \operatorname{div}(g\nabla\cdot)$ with Dirichlet boundary conditions. Expectation is $\mathbb{E}$, and P_f is the law of Y_n. In the white-noise model, by the Cameron–Martin theorem (see e.g. [6]),

$$\mathrm{KL}(P_f, P_g) = \tfrac{n}{2}\|u_f - u_g\|_{L^2}^2.$$

We use the following known results without proof.

(E1) *Global H^2 regularity* [5, Theorem 8.12], combined with the energy estimate. Let $g \in C^{0,1}(D)$ with $g \geq 1-r$. The solution $w \in H_0^1(D)$ of $\operatorname{div}(g\nabla w) = F \in L^2$ satisfies $\|w\|_{H^2} \leq C\|F\|_2$, where C depends only on d, $1-r$ and an upper bound for $\|g\|_{C^{0,1}}$.

(E2) *$W^{2,p}$ theory*, $1 < p < \infty$ [5, Theorem 9.15, Lemma 9.17]. Let $g \in C^1(D)$ with $g \geq 1-r$. The Dirichlet problem $g\Delta w + \nabla g\cdot\nabla w = F \in L^p$, $w \in W^{2,p} \cap W_0^{1,p}$, has a unique solution, and $\|w\|_{W^{2,p}} \leq C\|F\|_p$. The constant is uniform over $\mathcal{F}(\alpha, r)$, which is bounded in C^α and hence precompact in C^1.

(GN) *Gagliardo–Nirenberg inequality on the ball* [4, 8]. For $p > d$ and $t = (1 + d/p)/2$, $\|\nabla w\|_\infty \le C\|w\|_{W^{2,p}}^t \|w\|_p^{1-t} + C\|w\|_p$.

(As) *Assouad's lemma* [1]; see also [12, Ch. 2]. Let $\{P_\theta : \theta \in \{0,1\}^M\}$ be probability measures with $\mathrm{TV}(P_\theta, P_{\theta'}) \le \tau < 1$ whenever θ, θ' differ in exactly one coordinate. Then for every estimator $\hat\theta$ with values in $\{0,1\}^M$, $\max_\theta \mathbb{E}_\theta \sum_k \mathbf{1}\{\hat\theta_k \ne \theta_k\} \ge \frac{M}{2}(1-\tau)$. We also use Pinsker's inequality $\mathrm{TV} \le \sqrt{\mathrm{KL}/2}$ [12, Ch. 2].

(C1) *Seeley-type extension* [9]; see also [10]. There is a linear operator E with $Eg = g$ on D that is bounded $C^s(D) \to C^s(\mathbb{R}^d)$ for $0 < s \le S_0$, where we fix $S_0 > \alpha + 2$. It has the form $Eg(t) = \chi(|t|) \sum_i a_i\, g(R_i t)$ for $|t| > 1$, where $R_i t = (1 - \lambda_i(|t| - 1))\, t/|t|$, finitely many $\lambda_i > 0$, and χ is a smooth cutoff vanishing for $|t| \ge 1 + \eta$, with $\lambda_i \eta < 1$.

(C2) *Kernel approximation* (see e.g. [6]). Let $k \in C_c^\infty(\mathbb{R}^d)$ with $\int k = 1$ and vanishing moments of orders $1, \dots, N_0$, where we fix an integer $N_0 \ge 2\alpha + 2$. Put $k_J = 2^{Jd} k(2^J \cdot)$. Then $\|D^m(k_J * g - g)\|_\infty \le C 2^{-J(s-m)} \|g\|_{C^s}$ for integers $0 \le m < s$ and $0 < s \le N_0$.

(LP) *Littlewood–Paley theory* [10, 6]. For a fixed Littlewood–Paley decomposition $g = \sum_{j \ge 0} \Delta_j g$ we have $\|g\|_{C^s} \asymp \sup_j 2^{js} \|\Delta_j g\|_\infty$ for $s > 0$. Bernstein's inequality gives $\|D^m \Delta_j g\|_\infty \le C 2^{jm} \|\Delta_j g\|_\infty$. Moreover $C^s \subset C^{1,\beta_0}$ for $s > 1$, with $\beta_0 = \min(s - 1, 1/2)$.

(C3) *Taylor formula in Zygmund spaces; algebra property* (standard; see [10]). Let $\phi \in C^s$, $s > 0$, and $|h| \le 1$. Then $\Delta_h \phi(x) := \phi(x + h) - \phi(x) = P_\phi(x, h) + r_\phi(x, h)$, where $P_\phi(x,h) = \sum_{1 \le |\iota| < s} \partial^\iota \phi(x) h^\iota / \iota!$ and $|r_\phi(x,h)| \le C\|\phi\|_{C^s} |h|^s \Lambda(h)$ with $\Lambda(h) = 1 + \log(1/|h|)$; the logarithm is needed only when s is an integer. Also $C^s \cdot C^s \subset C^s$, with $\|\phi\psi\|_{C^s} \le C\|\phi\|_{C^s}\|\psi\|_{C^s}$.

(S) *Schauder estimate in Zygmund spaces* (standard). The Dirichlet Laplacian satisfies: $\Delta_D^{-1} : C^{\alpha-1}(D) \to C^{\alpha+1}(D)$ is bounded for $\alpha - 1 > 0$. For non-integer α this is classical Schauder theory [5, Ch. 6].

(ODE) C^k *dependence of flows on initial data*, with the variational equations [7, Ch. V].

(G) *Gaussian facts* [6]. Let $y \mapsto \varphi_y \in L^2(D)$ be C^{m+1} on an open set, with locally Lipschitz derivatives. Then $\mathbb{W}(\varphi_y)$ has a C^m version with $\partial_y^\iota \mathbb{W}(\varphi_y) = \mathbb{W}(\partial_y^\iota \varphi_y)$ for $|\iota| \le m$. Bochner integrals in L^2 commute with $\mathbb{W}$. We also use Dudley's entropy bound [3] and the Borell–Tsirelson–Ibragimov–Sudakov (Borell–TIS) inequality [2, 11].

5 The lower bound

Throughout this section $\alpha > 1$, $d \ge 2$ and $r \in (0,1)$ are fixed, and $j \ge j_0$ is large.

5.1 The hypercube

Directions and bumps. Choose $\omega_1, \dots, \omega_M \in S^{d-1}$ with pairwise geodesic distance $\ge 2 \cdot 2^{-j}$ and $M \ge c_d 2^{j(d-1)}$ (a maximal separated set). Fix $\varphi_0 \in C_c^\infty([0, 1/2))$ that is constant near 0 and not identically 0. Let $\Phi_k(\omega) = \varphi_0(2^j d_S(\omega, \omega_k))$, where d_S is the geodesic distance. The Φ_k are smooth, are rotations of one another, and have pairwise disjoint supports, which are caps of radius $2^{-j}/2$.

Extension and scaling. Extend each Φ_k to $\mathbb{R}^d \setminus \{0\}$ as a function homogeneous of degree 0. Then

$$\Delta\Phi_k(x) = |x|^{-2}(\Delta_S\Phi_k)(x/|x|),$$

where Δ_S is the Laplace–Beltrami operator. Put $\chi_k := 2^{-2j}\Delta_S\Phi_k$. In rescaled normal coordinates this is the flat Laplacian of a fixed nonzero compactly supported bump, up to $O(2^{-j})$ metric corrections. Hence $\|\partial^m\chi_k\|_\infty \le C_m 2^{jm}$, and for $j \ge j_0$

$$c2^{-j(d-1)} \le \|\chi_k\|^2_{L^2(S^{d-1})} \le C2^{-j(d-1)}.$$

The lower bound holds because the Laplacian of a nonzero compactly supported function is nonzero, and the value does not depend on k by rotation invariance.

The perturbations. Fix $b \in C_c^\infty((1/4, 3/4))$, not identically 0. Define

$$h_k(x) := \kappa 2^{-j(\alpha+2)} b(|x|)\,\Delta\Phi_k(x) = \kappa 2^{-j\alpha} b(|x|)|x|^{-2}\chi_k(x/|x|), \qquad f_\theta := \mathbf{1} + \sum_{k=1}^{M} \theta_k h_k, \quad \theta \in \{0,1\}^M.$$

Lemma 5.1. *There is $\kappa = \kappa(r) > 0$ such that $f_\theta \in \mathcal{F}(\alpha, r)$ for all θ and all j. The supports S_k of the h_k are pairwise disjoint, and*

$$\|h_k\|_2^2 \asymp 2^{-j(2\alpha+d-1)}, \qquad \|\Phi_k\|_{L^2(\{1/4<|x|<3/4\})} \asymp 2^{-j(d-1)/2}, \qquad \Big\|\sum_k \theta_k\Phi_k\Big\|_\infty \le \|\varphi_0\|_\infty.$$

Proof. Let $G_\theta = \sum_k \theta_k\chi_k$. Each point of the sphere has a neighbourhood meeting at most one support, so $\|D^m G_\theta\|_\infty \le C2^{jm}$ and $\|G_\theta\|_\infty \le C$. Take an integer $m > \alpha$. Then $|\Delta_h^m(2^{-j\alpha}G_\theta)| \le C2^{-j\alpha}\min((2^j|h|)^m, 1) \le C|h|^\alpha$. Hence $\|2^{-j\alpha}G_\theta\|_{C^\alpha(S^{d-1})} \le C$ uniformly in θ and j. Composing with $x \mapsto x/|x|$ and multiplying by the smooth function $b(|x|)|x|^{-2}$, supported in the annulus, keeps this bounded. Choosing κ small gives $\|f_\theta - \mathbf{1}\|_{C^\alpha(D)} \le r$. Disjointness of the supports is clear. In polar coordinates, $\|h_k\|_2^2 = \kappa^2 2^{-2j\alpha}\int b(\rho)^2\rho^{d-5}d\rho\,\|\chi_k\|^2_{L^2(S^{d-1})} \asymp 2^{-j(2\alpha+d-1)}$. The remaining two statements are immediate from the construction. □

By disjointness, $\|f_\theta - f_{\theta'}\|_2^2 = \sum_{k:\theta_k\neq\theta'_k}\|h_k\|_2^2$.

5.2 The key estimate

Proposition 5.2. *If $\theta, \theta' \in \{0,1\}^M$ differ only in coordinate k, then $\|u_{f_\theta} - u_{f_{\theta'}}\|_2 \le C2^{-2j}\|h_k\|_2$. Consequently $\|u_{f_\theta} - u_{f_{\theta'}}\|_2^2 \le C2^{-j(2\alpha+d+3)}$.*

Proof. Exact identity. Let $f = f_{\theta'}$, $g = f_\theta = f + h$ with $h = \pm h_k$, and $V = \nabla u_f$. Subtracting the two equations gives $\operatorname{div}(g\nabla(u_g - u_f)) = 1 - \operatorname{div}(g\nabla u_f) = -\operatorname{div}(hV)$. Hence

$$u_g - u_f = -L_g^{-1}\operatorname{div}(hV).$$

(i) Duality. L_g is self-adjoint on H_0^1, so $\|L_g^{-1}F\|_2 = \sup_{\|\phi\|_2\le 1}\langle F, \psi\rangle$ with $\psi := L_g^{-1}\phi$. Since $g \ge 1 - r$ and g is Lipschitz (as $\alpha > 1$), (E1) gives $\|\psi\|_{H^2} \le C$. It therefore suffices to show

$$|\langle \operatorname{div}(h_k V), \psi\rangle| \le C2^{-2j}\|h_k\|_2\|\psi\|_{H^2} \qquad \text{for } \psi \in H^2 \cap H_0^1. \tag{1}$$

Split $V = x/d + \nabla v$ with $v := u_f - u_1$.

(ii) Radial part. For $G = b(|x|)\Delta\Phi_k$ we claim

$$\operatorname{div}(G\,x/d) = b_1(|x|)\Delta\Phi_k, \qquad b_1(\rho) := \frac{\rho b'(\rho) + (d-2)b(\rho)}{d} \in C_c^\infty((1/4, 3/4)). \tag{2}$$

Indeed, $\Psi := \Delta\Phi_k$ is homogeneous of degree -2, so Euler's relation gives $x \cdot \nabla G = \rho b'\Psi - 2b\Psi$. Since $\operatorname{div}(G\,x) = x \cdot \nabla G + dG$, dividing by d gives (2). The function $b_1\psi$ has compact support in the annulus, where Φ_k is smooth. Integrating by parts twice and using Lemma 5.1,

$$|\langle \operatorname{div}(h_k\,x/d), \psi\rangle| = \kappa 2^{-j(\alpha+2)}\Big|\int \Phi_k\,\Delta(b_1\psi)\Big| \le C2^{-j(\alpha+2)}2^{-j(d-1)/2}\|\psi\|_{H^2} \le C2^{-2j}\|h_k\|_2\|\psi\|_{H^2}.$$

(iii) Drift part. Since h_k has compact support, $|\langle \operatorname{div}(h_k\nabla v), \psi\rangle| = |\int h_k\nabla v\cdot\nabla\psi| \le \|h_k\|_2\|\nabla v\|_\infty\|\psi\|_{H^1}$. It therefore suffices to show $\|\nabla v\|_\infty \le C2^{-j(\alpha+1-d/p)}$ for every $p > d$, and then to take $p > d$ with $d/p \le \alpha - 1$.

Equation for v. We have $\operatorname{div}(f\nabla v) = 1 - \operatorname{div}(f\,x/d) = -\operatorname{div}((f-1)x/d) =: -q$. By (2) applied termwise, $q = \kappa 2^{-j(\alpha+2)}b_1(|x|)\Delta\Phi_{\theta'}$ with $\Phi_{\theta'} = \sum_l \theta'_l\Phi_l$.

$W^{2,p}$ bound. Since $q = \kappa 2^{-j\alpha}b_1(|x|)|x|^{-2}\sum_l \theta'_l\chi_l(x/|x|)$ and the χ_l are bounded with disjoint supports, $\|q\|_p \le C2^{-j\alpha}$. Hence $\|v\|_{W^{2,p}} \le C2^{-j\alpha}$ by (E2).

L^p bound. Take ϕ with $\|\phi\|_{p'} \le 1$ and let $\psi = L_f^{-1}\phi \in W^{2,p'} \cap W_0^{1,p'}$, so that $\|\psi\|_{W^{2,p'}} \le C$ by (E2). By Green's identity, and integrating by parts twice as in (ii),

$$\Big|\int v\phi\Big| = \Big|\int q\psi\Big| = \kappa 2^{-j(\alpha+2)}\Big|\int \Phi_{\theta'}\,\Delta(b_1\psi)\Big| \le C2^{-j(\alpha+2)}\|\varphi_0\|_\infty\|\psi\|_{W^{2,p'}} \le C2^{-j(\alpha+2)}.$$

So $\|v\|_p \le C2^{-j(\alpha+2)}$.

Interpolation. By (GN), $\|\nabla v\|_\infty \le C2^{-j\alpha t}2^{-j(\alpha+2)(1-t)} + C2^{-j(\alpha+2)} \le C2^{-j(\alpha+1-d/p)}$, since $2(1-t) = 1 - d/p$.

Combining (ii) and (iii) proves (1), hence the first claim. The second follows from $\|h_k\|_2^2 \le C2^{-j(2\alpha+d-1)}$. □

5.3 Conclusion via Assouad's lemma

Proof of Theorem 2.1(i). By Proposition 5.2, $\mathrm{KL}(P_{f_\theta}, P_{f_{\theta'}}) \le Cn2^{-j(2\alpha+d+3)}$ for neighbours θ, θ'. Choose $2^j \asymp n^{1/(2\alpha+d+3)}$, with a constant large enough that $\mathrm{KL} \le 1/8$ for all neighbours. Pinsker's inequality then gives $\mathrm{TV} \le 1/4$.

Given any estimator $\hat f$, let $\hat\theta_k$ minimise $\|(\hat f - 1 - th_k)\mathbf{1}_{S_k}\|_2$ over $t \in \{0, 1\}$. On S_k we have $f_\theta = 1 + \theta_k h_k$. If $\hat\theta_k \ne \theta_k$, the triangle inequality gives $\|h_k\|_2 \le \|(\hat f - 1 - \hat\theta_k h_k)\mathbf{1}_{S_k}\|_2 + \|(\hat f - 1 - \theta_k h_k)\mathbf{1}_{S_k}\|_2 \le 2\|(\hat f - f_\theta)\mathbf{1}_{S_k}\|_2$. Since the S_k are disjoint,

$$\|\hat f - f_\theta\|_2^2 \ge \tfrac14 \sum_k \|h_k\|_2^2\,\mathbf{1}\{\hat\theta_k \ne \theta_k\}.$$

By (As) with $\tau = 1/4$ and Lemma 5.1,

$$\sup_{f\in\mathcal{F}(\alpha,r)} \mathbb{E}_f\|\hat f - f\|_2^2 \ge \max_\theta \mathbb{E}_\theta\|\hat f - f_\theta\|_2^2 \ge \tfrac14 \min_k \|h_k\|_2^2 \cdot \tfrac{M}{2} \cdot \tfrac34 \ge c2^{j(d-1)}2^{-j(2\alpha+d-1)} = c2^{-2j\alpha} \asymp n^{-2\alpha/(2\alpha+d+3)}.$$

The condition $\alpha > 1$ was used only in (iii), and to make $f \in C^1$ for (E1) and (E2). □

6 The upper bound

Throughout, $\alpha > 1$ and $d \geq 2$ are fixed, $f \in \mathcal{F}(\alpha, r)$, and $r \leq r_0$, where r_0 is reduced finitely many times below.

6.1 A Schauder bound

Lemma 6.1. *There is $r_0 = r_0(d, \alpha)$ such that for $r \leq r_0$ and all $f \in \mathcal{F}(\alpha, r)$,*

$$\|u_f - u_1\|_{C^{\alpha+1}(D)} \leq Cr, \qquad \|u_f\|_{C^{\alpha+1}(D)} \leq C. \tag{3}$$

Proof. Write $L_f = \Delta + \operatorname{div}((f-1)\nabla\cdot)$. By the algebra property (C3), $\|\operatorname{div}((f-1)\nabla v)\|_{C^{\alpha-1}} \leq C_1\|f-1\|_{C^\alpha}\|v\|_{C^{\alpha+1}} \leq C_1 r\|v\|_{C^{\alpha+1}}$. Take r_0 with $C_1 r_0 \|\Delta_D^{-1}\| \leq 1/2$. Writing $L_f v = F$ as $(I + \Delta_D^{-1}\operatorname{div}((f-1)\nabla\cdot))v = \Delta_D^{-1}F$ on the closed subspace of $C^{\alpha+1}(D)$ with zero trace, a Neumann series using (S) shows that $L_f^{-1} : C^{\alpha-1}(D) \to C^{\alpha+1}(D)$ (Dirichlet data) has norm $\leq 2\|\Delta_D^{-1}\|$. The solution so obtained is classical, hence equal to the weak solution. As in Proposition 5.2, $u_f - u_1 = -L_f^{-1}\operatorname{div}((f-1)x/d)$, and the right-hand side has $C^{\alpha-1}$ norm $\leq Cr$. □

This is the first place where smallness of r enters.

6.2 Smoothing, noise and the estimator

Resolution and smoother. Let $J := \lfloor \log_2 n/(2\alpha + d + 3)\rfloor$. Then

$$n^{-1/2} \leq 2^{-J(\alpha+(d+3)/2)}, \qquad n^{-1}2^{J(d+3)} \leq 2^{-2J\alpha}, \qquad 2^{-2J\alpha} \leq 2^{2\alpha}n^{-2\alpha/(2\alpha+d+3)}, \qquad J \leq \log_2 n. \tag{4}$$

Put $K_J g := (k_J * Eg)|_D$, with k_J from (C2) and E from (C1). On data, $K_J Y(x) = \langle Y, k_J(x,\cdot)\rangle$, where $k_J(x,\cdot) := E^*[k_J(x-\cdot)] \in L^2(D)$; this is defined for all $x \in \mathbb{R}^d$. By the change of variables $z = R_i t$, E^* is the identity on D plus finitely many terms of the form $a_i(\chi\varphi)(R_i^{-1}z)|\det DR_i^{-1}(z)|$. These are supported on shells $\{1 - \lambda_i\eta < |z| < 1\}$ and have bounded Jacobians. This gives the following kernel bounds.

(K1) $|\partial_x^\iota k_J(x,z)| \leq C2^{J(d+|\iota|)}$ for $|\iota| \leq 4$.

(K2) $\partial_x^\iota k_J(x,z) = 0$ unless $|x-z| \leq C_0 2^{-J}$. Indeed, for the reflected terms let $t = R_i^{-1}z$, so $|t| > 1$. If $x \in D$ and $|x-t| \leq C2^{-J}$, then $|t|-1 \leq C2^{-J}$ and $|z-t| = (1+\lambda_i)(|t|-1) \leq C2^{-J}$.

(K3) $\|\partial_x^\iota k_J(x,\cdot)\|_2^2 \leq C2^{J(d+2|\iota|)}$, and $x \mapsto \partial_x^\iota k_J(x,\cdot)$ is Lipschitz into L^2 with constant $C2^{J(d/2+|\iota|+1)}$, for $|\iota| \leq 3$. This follows from (K1), (K2) and the fact that the support in z has volume $\leq C2^{-Jd}$.

Notation. Let $U = Eu$, $F = Ef$, $V = \nabla u$ and $V^E = \nabla U$. By (C1) and (3), $\|F\|_{C^\alpha} + \|V^E\|_{C^\alpha} \leq C$. Let

$$V_J := \nabla K_J u = (k_J * V^E)|_D, \qquad e := K_J u - u, \qquad \xi := n^{-1/2}K_J\mathbb{W}, \qquad \hat{V} := \nabla K_J Y = V_J + \nabla\xi.$$

V_J is the restriction of a C^∞ field on $\mathbb{R}^d$. By (G) and (K1)–(K3), ξ is a centred Gaussian field with C^3 sample paths on a neighbourhood of D (it is in fact smooth), and ξ does not depend on f.

Good event. Let

$$\mathcal{A} := \Big\{ \max_{|\iota|=m} \sup_D |\partial^\iota \xi| \le J n^{-1/2} 2^{J(d/2+m)},\ m = 1,2,3 \Big\}.$$

On $\mathcal{A}$, by (4) and after adjusting constants for operator norms,

$$\|\nabla \xi\|_\infty \le CJ2^{-J(\alpha+1/2)}, \qquad \|D^2\xi\|_\infty \le CJ2^{-J(\alpha-1/2)}, \qquad \|D^3\xi\|_\infty \le CJ2^{-J(\alpha-3/2)}. \tag{5}$$

Definition 6.2 (Estimator)**.** Let $\varepsilon_0 := 1/(8d)$, and define the observable event $\mathcal{B} := \{\|\hat{V} - x/d\|_{C^1(D)} \le \varepsilon_0\}$. On $\mathcal{B}$ set $\hat{f} := T_{\hat{V}}^{-1}\mathbf{1}$ (Definition 6.6); off $\mathcal{B}$ set $\hat{f} := \mathbf{1}$. Let $\tilde{f} := \min(\max(\hat{f}, 1-r), 1+r)$.

Since $\|f - 1\|_\infty \le r$, clipping can only decrease $|\tilde{f} - f|$ pointwise. The estimator uses only Y, n, α, r and the fixed kernels. The event $\mathcal{A}$ itself is not observable: ξ does not depend on f, but $\xi = K_J Y - K_J u$ cannot be computed from the data because u is unknown. This is why the estimator is defined on $\mathcal{B}$; Lemma 6.4 shows $\mathcal{A} \subset \mathcal{B}$.

6.3 The smoothed fields

Lemma 6.3. *Let $g \in C^s(\mathbb{R}^d)$, $s > 0$, and let $m \ge 1$ be an integer. Then $\|D^m(k_J * g)\|_\infty$ is at most $C2^{J(m-s)}\|g\|_{C^s}$ if $m > s$, at most $C(J+1)\|g\|_{C^s}$ if $m = s$, and at most $C\|g\|_{C^s}$ if $m < s$.*

Proof. By (LP), $D^m(k_J * g) = \sum_j (D^m k_J) * \Delta_j g$, and the series converges uniformly because each term is $\le \|D^m k_J\|_1 \|\Delta_j g\|_\infty \le C2^{Jm}2^{-js}\|g\|_{C^s}$. For $j \le J$, write the term as $k_J * D^m \Delta_j g$; by Bernstein it is $\le C2^{j(m-s)}\|g\|_{C^s}$. For $j > J$, the first bound gives $C2^{Jm-js}\|g\|_{C^s}$, and summing over $j > J$ gives $C2^{J(m-s)}\|g\|_{C^s}$. Summing the bound $C2^{j(m-s)}$ over $j \le J$ gives the three cases. □

Lemma 6.4. *Let $\varepsilon \le \varepsilon_0$. There are r_0 and J_0 such that for $r \le r_0$ and $J \ge J_0$:*

(1) $\|V_J - x/d\|_{C^1(D)} \le \varepsilon/2$;

(2) on $\mathcal{A}$, $\|\hat{V} - x/d\|_{C^1(D)} \le \varepsilon$; in particular (with $\varepsilon = \varepsilon_0$) $\mathcal{A} \subset \mathcal{B}$;

(3) $\|D^m V_J\|_\infty \le CJ2^{J(m-\alpha)_+}$ for $m = 2, 3$, with no factor J unless $\alpha = m$.

Proof. Item (3) is Lemma 6.3 with $g = V^E$ and $s = \alpha$. For (1), on D we have $V_J - x/d = (k_J * V^E - V^E) + \nabla(u - u_1)$. By (LP), $D^i V^E$ is β_0-Hölder for $i = 0, 1$, with $\beta_0 = \min(\alpha - 1, 1/2)$. Hence $|D^i(k_J * V^E - V^E)| \le C \int |k_J(y)||y|^{\beta_0} dy \le C2^{-J\beta_0}$. By (3), $\|\nabla(u - u_1)\|_{C^1} \le Cr$. Take r_0 and J_0 such that $C(2^{-J_0\beta_0} + r_0) \le \varepsilon/2$. For (2), on $\mathcal{A}$, (5) gives $\|\nabla\xi\|_{C^1} \le CJ2^{-J(\alpha-1/2)} \le \varepsilon/2$ for $J \ge J_0$. □

6.4 Geometry of the flow and the transport inverse

Setting. Let X be the restriction to D of a C^k field ($k \ge 1$) on a neighbourhood of D, with

$$\|X - x/d\|_{C^1(D)} := \max\big(\sup_D |X - x/d|,\ \sup_D \|DX - I/d\|\big) \le \varepsilon \le 1/(8d).$$

Put $\nu := 1/d - \varepsilon \ge 7/(8d)$. Let Φ_t be the flow of X, let $\gamma(s) = \gamma_x(s) = \Phi_{-s}x$ be the backward ray from x, and let

$$w(s,x) := \exp\Big(- \int_0^s \operatorname{div} X(\Phi_{-\sigma}x)\, d\sigma \Big).$$

Lemma 6.5. *In this setting the following hold.*

(1) X has exactly one zero $x_X \in D$, and $|x_X| \le d\varepsilon$.

(2) With $\rho(y) = |y - x_X|$: $(y - x_X) \cdot X(y) \ge (1/d - \varepsilon)\rho^2$ and $|X(y)| \le (1/d + \varepsilon)\rho$ on D.

(3) For $x \in D$, $\Phi_{-s}x$ exists for all $s \ge 0$ and stays in D. It lies in D° for $s > 0$, and D° is invariant.

(4) $\varrho(s) := \rho(\gamma(s))$ satisfies $\varrho(0)e^{-(1/d+\varepsilon)s} \le \varrho(s) \le \varrho(0)e^{-(1/d-\varepsilon)s}$. If $\varrho(0) > 0$, then $-(1/d + \varepsilon)\varrho \le \varrho' \le -\nu\varrho$.

(5) The propagator $P(s, \sigma)$ of $y' = -DX(\gamma)y$ satisfies $\|P(s, \sigma)\| \le e^{-\nu(s-\sigma)}$. In particular $\|D\Phi_{-s}\| \le e^{-\nu s}$.

(6) $1 - d\varepsilon \le \operatorname{div} X \le 1 + d\varepsilon$, so $e^{-(1+d\varepsilon)s} \le w \le e^{-(1-d\varepsilon)s} \le e^{-7s/8}$.

Proof. (1) The map $x \mapsto x - dX(x) = -d(X - x/d)$ sends D into $B(0, d\varepsilon) \subset D$ and is a $d\varepsilon$-contraction on the convex set D. Its unique fixed point is the unique zero of X. (2) $X(y) = Q(y)(y - x_X)$ with $Q = \int_0^1 DX(x_X + t(y - x_X))\,dt$ and $\|Q - I/d\| \le \varepsilon$; the segment lies in D by convexity. (3) For $|z| = 1$, $z \cdot X(z) \ge 1/d - \varepsilon > 0$, so $\frac{d}{ds}|\gamma|^2 = -2\gamma \cdot X(\gamma) < 0$ whenever $|\gamma| = 1$. A first-exit-time argument gives invariance of D and of D°, and shows that boundary points enter D° immediately. Global existence holds because D is compact. (4) $(\varrho^2)' = -2(\gamma - x_X) \cdot X(\gamma)$; apply (2) and Gronwall's inequality. (5) $v \cdot DX\, v \ge \nu|v|^2$ on D, so $(|y|^2)' \le -2\nu|y|^2$. Also $D\Phi_{-s} = P(s, 0)$. (6) $\operatorname{div} X = 1 + \operatorname{tr}(DX - I/d)$ and $|\operatorname{tr} A| \le d\|A\|$. □

Definition 6.6. For $g \in C^1(D)$ put $T_X g := \operatorname{div}(gX)$. For $H \in C(D)$ put

$$T_X^{-1}H(x) := \int_0^\infty H(\Phi_{-s}x)\, w(s, x)\, ds.$$

By Lemma 6.5(6) the integral converges absolutely and uniformly, and $T_X^{-1}H \in C(D)$.

Since $w(s, \Phi_{-t}x) = w(s + t, x)/w(t, x)$, we have the semigroup identity

$$w(t, x)\, T_X^{-1}H(\Phi_{-t}x) = \int_t^\infty H(\Phi_{-s}x)\, w(s, x)\, ds. \tag{6}$$

Lemma 6.7. *In the setting of Lemma 6.5:*

(a) (Uniqueness.) *If $g \in C^1(D)$ and $\operatorname{div}(gX) = H$ on D° with $H \in C(D)$, then $g = T_X^{-1}H$.*

(a′) (Right inverse.) *If $H \in C(D)$ and $g := T_X^{-1}H \in C^1(D)$, then $\operatorname{div}(gX) = H$ on D.*

(b) $\|T_X^{-1}H\|_\infty \le \frac{8}{7}\|H\|_\infty \le 2\|H\|_\infty$.

(c) (Boundary layer.) *If $H = 0$ on $\{|x| \le 1 - \delta\}$, then $\|T_X^{-1}H\|_\infty \le 6d\delta\|H\|_\infty$.*

Proof. (a) Fix $x \in D$. For $s > 0$, $\gamma(s) \in D^\circ$, and $\psi(s) = g(\gamma(s))w(s, x)$ satisfies $\psi' = -(X \cdot \nabla g + g \operatorname{div} X)(\gamma)\, w = -H(\gamma)\, w$. Integrate over $[\delta, S]$, then let $\delta \to 0$ (by continuity) and $S \to \infty$ (using $|\psi(S)| \le \|g\|_\infty e^{-7S/8}$). This gives $g(x) = \int_0^\infty H(\gamma(s))w(s, x)\, ds$.

(a′) Fix $x \in D^\circ$ and differentiate (6) at $t = 0^+$. The right side gives $-H(x)$. Since $w(0, x) = 1$ and $\partial_t w(0, x) = -\operatorname{div} X(x)$, the left side gives $-\nabla g(x) \cdot X(x) - g(x) \operatorname{div} X(x)$. Extend to D by continuity.

(b) Use $w \le e^{-7s/8}$.

(c) The case $\delta > 1/2$ follows from (b), since then $6d\delta > 2$. Let $\delta \le 1/2$ and $\phi = |\gamma|^2$. Where $|\gamma| \ge 1/2$, $\gamma \cdot X(\gamma) \ge |\gamma|(|\gamma|/d - \varepsilon) \ge 3/(16d)$, so $\phi' \le -3/(8d)$. Hence ϕ drops below $(1-\delta)^2$ within time $\le (1-(1-\delta)^2) \cdot 8d/3 \le 6d\delta$, and by the same sign argument it never returns. Since $H(\gamma(s)) = 0$ afterwards and $w \le 1$, the claim follows. □

Lemma 6.8 (Flow derivatives, uniform in s)**.** *Let $M = D\Phi_{-s}$, $N = D^2\Phi_{-s}$ and $\ell = \log w$, all differentiated in x.*

(1) If $X \in C^2$: $\|N(s)\| \le \|D^2 X\|_\infty e^{-\nu s}/\nu$ and $|D_x \ell| \le \|D \operatorname{div} X\|_\infty/\nu$.

(2) If $X \in C^3$: $\|D_x^2 \ell\| \le \|D^2 \operatorname{div} X\|_\infty/(2\nu) + \|D \operatorname{div} X\|_\infty \|D^2 X\|_\infty/\nu^2$.

Proof. N solves $N' = -DX(\gamma)N - D^2X(\gamma)[M\cdot, M\cdot]$ with $N(0) = 0$. By Duhamel's formula and Lemma 6.5(5), $\|N(s)\| \le \int_0^s e^{-\nu(s-\sigma)} \|D^2X\|_\infty e^{-2\nu\sigma} d\sigma \le \|D^2X\|_\infty e^{-\nu s}/\nu$. Next, $D_x\ell[v] = -\int_0^s D \operatorname{div} X(\gamma)[Mv]\, d\sigma$, and $\int_0^\infty e^{-\nu\sigma} d\sigma = 1/\nu$. Finally, $D_x^2\ell[v,v'] = -\int_0^s \big(D^2 \operatorname{div} X(\gamma)[Mv, Mv'] + D \operatorname{div} X(\gamma)[N[v,v']]\big) d\sigma$; bound this with the two previous estimates. □

Differentiation under the integral. Let the integrand be $H(\Phi_{-s}x)w(s,x)$ or $w(s,x)$. By (ODE), and since the flow is defined on a neighbourhood of D, its x-derivatives up to the relevant order are jointly continuous on $[0,\infty) \times D$. By Lemma 6.8 and the identities $D_x w = w\, D_x\ell$ and $D_x^2 w = w(D_x^2\ell + D_x\ell \otimes D_x\ell)$, they are dominated by $Ke^{-7s/8}$. Truncating the integral at S, differentiating, and passing to the uniform limit gives the following.

Lemma 6.9. *All constants below depend only on d, uniformly over $\|X - x/d\|_{C^1} \le \varepsilon \le 1/(8d)$.*

(d) *If $X \in C^2$ and $H \in C^1(D)$, then $T_X^{-1}H \in C^1(D)$ and $\|\nabla T_X^{-1} H\|_\infty \le C_d(\|\nabla H\|_\infty + \|H\|_\infty \|D \operatorname{div} X\|_\infty)$.*

(e) *If $X \in C^3$, then $T_X^{-1}\mathbf{1} \in C^2(D)$, $\|\nabla T_X^{-1}\mathbf{1}\|_\infty \le C_d \|D \operatorname{div} X\|_\infty$, and $\|D^2 T_X^{-1}\mathbf{1}\|_\infty \le C_d\big(\|D^2 \operatorname{div} X\|_\infty + \|D \operatorname{div} X\|_\infty^2 + \|D \operatorname{div} X\|_\infty \|D^2 X\|_\infty\big)$.*

6.5 The centring function and an exact decomposition

Fix $\varepsilon = \varepsilon_0$ in Lemma 6.4, so that Lemmas 6.5–6.9 apply to $X = V_J$ (with $\varepsilon_0/2$), and on $\mathcal{A}$ also to $X = \hat{V}$. Let

$$\bar{f} := T_{V_J}^{-1}\mathbf{1}.$$

Using $\|D^m \operatorname{div} V_J\|_\infty \le d\|D^{m+1}V_J\|_\infty$, Lemma 6.4(3) and Lemma 6.9(e),

$$\|\bar{f}\|_\infty \le 2, \qquad \|\nabla \bar{f}\|_\infty \le CJ2^{J(2-\alpha)_+}, \qquad \|D^2 \bar{f}\|_\infty \le CJ^2 2^{J(3-\alpha)_+}. \tag{7}$$

The last bound uses $2(2-\alpha)_+ \le (3-\alpha)_+$ for $\alpha \ge 1$. Moreover $\bar{f} \in C^2(D)$, and $\operatorname{div}(\bar{f}V_J) = 1$ by Lemma 6.7(a′).

Noise term. Let $G_\xi := \operatorname{div}(\bar{f}\nabla\xi) = \bar{f}\Delta\xi + \nabla\bar{f} \cdot \nabla\xi \in C^1(D)$ and $Z := T_{V_J}^{-1} G_\xi$. Since V_J is deterministic, Z is defined for every realisation. By Lemma 6.9(d), $Z \in C^1(D)$, and $T_{V_J} Z = G_\xi$ by Lemma 6.7(a′).

Proposition 6.10. *On $\mathcal{A}$ (so that $\hat{V}$ satisfies the hypotheses of Lemma 6.5 and $\hat{f} = T_{\hat{V}}^{-1}\mathbf{1}$),*

$$\hat{f} - f = (\bar{f} - f) - Z + T_{\hat{V}}^{-1} \operatorname{div}(Z\nabla\xi). \tag{8}$$

Proof. $\hat{V}$ is C^3 on a neighbourhood of D, so $\hat{f} \in C^2$ by Lemma 6.9(e), and $T_{\hat{V}}\hat{f} = 1$ by Lemma 6.7(a′). Hence $T_{\hat{V}}(\hat{f}-\bar{f}) = 1-\operatorname{div}(\bar{f}V_J)-\operatorname{div}(\bar{f}\nabla\xi) = -G_\xi$. Also $T_{\hat{V}}Z = T_{V_J}Z+\operatorname{div}(Z\nabla\xi) = G_\xi+\operatorname{div}(Z\nabla\xi)$. Therefore $T_{\hat{V}}(\hat{f}-\bar{f}+Z) = \operatorname{div}(Z\nabla\xi)$, which is continuous. Since $\hat{f}-\bar{f}+Z \in C^1$, Lemma 6.7(a) with $X = \hat{V}$ gives $\hat{f}-\bar{f}+Z = T_{\hat{V}}^{-1}\operatorname{div}(Z\nabla\xi)$, which is (8). □

6.6 The bias: $\|\bar{f}-f\|_\infty \le CJ^2 2^{-J\alpha}$

Proposition 6.11. *Uniformly over $f \in \mathcal{F}(\alpha, r)$, $\|\bar{f}-f\|_\infty \le CJ^2 2^{-J\alpha}$, with no factor J^2 when $\alpha \notin \mathbb{N}$.*

Proof. Reduction. $f \in C^1$ (as $\alpha > 1$), $u \in C^{\alpha+1} \subset C^2$ by (3), $\operatorname{div}(fV) = 1$, and $V_J - V = \nabla e$ on D. So $T_{V_J}(\bar{f}-f) = 1-\operatorname{div}(fV_J) = -\operatorname{div}(f\nabla e)$, and by Lemma 6.7(a), $\bar{f}-f = -T_{V_J}^{-1}\operatorname{div}(f\nabla e)$.

Commutator. Let $g := F - k_J * F$ and $R(x) := -\int k_J(-h)\Delta_h F(x)\Delta_h V^E(x)\,dh$. Then

$$F\,(k_J * V^E) = k_J * (FV^E) + gV^E + R.$$

Indeed, $F(x)(k_J * V^E)(x) - k_J * (FV^E)(x) = \int k_J(y)[F(x)-F(x-y)]V^E(x-y)\,dy$. Insert $F(x)-F(x-y) = -\Delta_{-y}F(x)$ and $V^E(x-y) = V^E(x)+\Delta_{-y}V^E(x)$, and use $\int k_J = 1$; the first part gives gV^E and the second, after $h = -y$, gives R. Let $\Gamma := \operatorname{div}(FV^E) - 1$, which lies in $C^{\alpha-1}(\mathbb{R}^d)$ by the algebra property and vanishes on D. Since $\operatorname{div}(k_J * (FV^E)) = k_J * \Gamma + 1$, on D

$$\operatorname{div}(f\nabla e) = \operatorname{div}(fV_J) - 1 = k_J * \Gamma + \operatorname{div}(gV) + \operatorname{div} R.$$

We treat the three terms in turn, using linearity of $T_{V_J}^{-1}$.

(i) The term $\operatorname{div}(gV)$. Since $V = V_J - \nabla e$, we have $\operatorname{div}(gV) = T_{V_J}g - \operatorname{div}(g\nabla e)$. As $g \in C^1$, Lemma 6.7(a) gives $T_{V_J}^{-1}\operatorname{div}(gV) = g - T_{V_J}^{-1}\operatorname{div}(g\nabla e)$. By (C2) (applied to $F \in C^\alpha$ and $U \in C^{\alpha+1}$): $\|g\|_\infty \le C2^{-J\alpha}$, $\|\nabla g\|_\infty \le C2^{-J(\alpha-1)}$, $\|\nabla e\|_\infty \le C2^{-J\alpha}$ and $\|D^2 e\|_\infty \le C2^{-J(\alpha-1)}$. By Lemma 6.7(b), the contribution is $\le C2^{-J\alpha} + C2^{-J(2\alpha-1)}$.

(ii) The term $\operatorname{div} R$. We have $\partial R = -\int k_J(-h)\big[\Delta_h\partial F\,\Delta_h V^E + \Delta_h F\,\Delta_h\partial V^E\big]dh$. Each product has the form $\Delta_h\phi_1\,\Delta_h\phi_2$ with $\phi_i \in C^{s_i}$ and $(s_1, s_2) = (\alpha-1, \alpha)$ or $(\alpha, \alpha-1)$. By (C3), and using $|P_{\phi_i}| \le C|h|$ for $|h| \le 1$, the product equals a polynomial in h (with coefficients depending on x) whose monomials have degrees in $[2, s_1+s_2) = [2, 2\alpha-1)$. The remainder is bounded by $C|h|^{\min(1+s_1,\,1+s_2,\,s_1+s_2)}\Lambda(h)^2 = C|h|^\alpha\Lambda(h)^2$. When $s_i < 1$, P_{ϕ_i} is empty. The polynomial terms integrate to 0 against $k_J(-h)$, because $N_0 \ge 2\alpha+2$. Since k_J is supported in $|h| \le C2^{-J}$, the substitution $h = 2^{-J}y$ gives $\int |k_J(h)||h|^\alpha\Lambda(h)^2 dh \le CJ^2 2^{-J\alpha}$. So $\|\operatorname{div} R\|_\infty \le CJ^2 2^{-J\alpha}$, with no J^2 for non-integer α (then no logarithm appears in (C3)). By Lemma 6.7(b), the contribution is $\le CJ^2 2^{-J\alpha}$.

(iii) The term $k_J * \Gamma$. On D, $k_J * \Gamma = k_J * \Gamma - \Gamma$, so by (C2) with $s = \alpha-1$ and $m = 0$ it is $O(2^{-J(\alpha-1)})$. By the support of k_J, it vanishes at points of D at distance $> C2^{-J}$ from ∂D. By Lemma 6.7(c) with $\delta = C2^{-J}$, the contribution is $\le C2^{-J}2^{-J(\alpha-1)} = C2^{-J\alpha}$.

Adding (i)–(iii) gives the claim. □

6.7 The variance: $\mathbb{E}\|Z\|_2^2 \le Cn^{-1}2^{J(d+3)} \le C2^{-2J\alpha}$

Proposition 6.12. $\mathbb{E}\|Z\|_{L^2(D)}^2 \le Cn^{-1}2^{J(d+3)} \le C2^{-2J\alpha}$.

Proof. Representation. Let $\mathcal{D}_y := \bar{f}(y)\Delta_y + \nabla\bar{f}(y)\cdot\nabla_y$. By (G), $G_\xi(y) = n^{-1/2}\mathbb{W}(\mathcal{D}_y k_J(y,\cdot))$, and since Bochner integrals commute with $\mathbb{W}$,

$$Z(x) = n^{-1/2}\mathbb{W}(h_x), \qquad h_x = \int_0^\infty w(s,x)\,\mathcal{D}_y k_J(y,\cdot)\big|_{y=\gamma_x(s)}\,ds,$$

where γ_x is the backward ray of V_J. Hence $\mathbb{E}Z(x)^2 = n^{-1}\|h_x\|_2^2$.

Kernel bound. Since $(2-\alpha)_+ < 1$, (7) gives $\|\nabla \bar f\|_\infty \le CJ2^{J(2-\alpha)_+} \le C2^J$. With (K1) and (K2), $|\mathcal{D}_y k_J(y,z)| \le C2^{J(d+2)}\mathbf{1}\{|y-z| \le \delta_J\}$, where $\delta_J := C_0 2^{-J}$. Let $\mu_x := \int_0^\infty w(s,x)\,\delta_{\gamma_x(s)}\,ds$, a measure on D of mass $\le 8/7$. Then

$$|h_x(z)| \le C2^{J(d+2)}\mu_x(B(z,\delta_J)), \qquad \|h_x\|_2^2 \le C2^{2J(d+2)}\int \mu_x(B(z,\delta_J))^2 dz \le C2^{2J(d+2)}\delta_J^d \int \mu_x(B(y,2\delta_J))\,\mu_x(dy).$$

The last step is Fubini: $\int \mu(B(z,\delta))^2 dz = \iint |B(y,\delta)\cap B(y',\delta)|\,\mu(dy)\mu(dy')$, and the intersection is empty unless $|y-y'| \le 2\delta$.

Ray measure. Let x_J be the zero of V_J, $\rho = |\cdot - x_J|$, and $\rho_x = \rho(x) > 0$. Let $\varrho(s) = \rho(\gamma_x(s))$. By Lemma 6.5(4),(6), $w(s,x) \le (\varrho(s)/\rho_x)^\beta$ with $\beta = d(1-d\varepsilon)/(1+d\varepsilon) \ge 7d/9 > 1$. Moreover ϱ decreases strictly from ρ_x to 0, with $|\varrho'| \ge \nu\varrho$. Changing variables $t = \varrho(s)$, for every $\varphi \ge 0$,

$$\int \varphi\, d\mu_x \le \frac{1}{\nu}\int_0^{\rho_x} \varphi(\gamma_x \text{ at radius } t)\,(t/\rho_x)^\beta\,\frac{dt}{t}. \tag{9}$$

Small-ball mass. If $|\gamma_x(s) - y| \le 2\delta_J$, then $|\varrho(s) - \rho(y)| \le 2\delta_J$. Hence, by (9), $m(y) := \mu_x(B(y,2\delta_J))$ satisfies

$$m(y) \le C\delta_J \rho(y)^{\beta-1}\rho_x^{-\beta} \text{ if } \rho(y) \ge 4\delta_J, \qquad m(y) \le C\min(1,\delta_J/\rho_x)^\beta \text{ if } \rho(y) < 4\delta_J.$$

(In the first case $t \in [\rho(y)/2, 3\rho(y)/2]$ ranges over an interval of length $4\delta_J$; in the second $t \le \min(6\delta_J, \rho_x)$.)

Integration. Apply (9) once more, with $\varphi = m$. The part with $t \ge 4\delta_J$ is $\le C\delta_J\rho_x^{-2\beta}\int_0^{\rho_x} t^{2\beta-2}dt \le C\delta_J/\rho_x$, since $2\beta - 1 > 0$. The part with $t < 4\delta_J$ is $\le C\min(1,\delta_J/\rho_x)^{2\beta} \le C\min(1,\delta_J/\rho_x)$. Trivially, $\int m\,d\mu_x \le \mu_x(D)^2 \le 4$. Hence $\int m\,d\mu_x \le C\min(1,\delta_J/\rho_x)$, and

$$\|h_x\|_2^2 \le C2^{J(d+4)}\min(1, 2^{-J}/\rho_x) \le C2^{J(d+3)}|x - x_J|^{-1}.$$

Since $\int_D |x-x_J|^{-1}dx \le C$ for $d \ge 2$, Tonelli's theorem and (4) give $\mathbb{E}\|Z\|_2^2 \le Cn^{-1}2^{J(d+3)} \le C2^{-2J\alpha}$. □

6.8 The remainder and the bad event

Lemma 6.13. *On $\mathcal{A}$, $\|T_{\hat V}^{-1}\operatorname{div}(Z\nabla\xi)\|_\infty \le CJ^4 2^{-J(2\alpha-1)}$. This is $o(2^{-J\alpha})$ because $\alpha > 1$.*

Proof. We use (5), (7), $(2-\alpha)_+ < 1$ and $(3-\alpha)_+ \le 2$. First,

$$\|G_\xi\|_\infty \le \|\bar f\|_\infty\|\Delta\xi\|_\infty + \|\nabla\bar f\|_\infty\|\nabla\xi\|_\infty \le CJ2^{-J(\alpha-1/2)} + CJ^2 2^{J(2-\alpha)_+ - J(\alpha+1/2)} \le CJ^2 2^{-J(\alpha-1/2)}.$$

Next, $\|\nabla G_\xi\|_\infty \le C(\|D^3\xi\|_\infty + \|\nabla\bar f\|_\infty\|D^2\xi\|_\infty + \|D^2\bar f\|_\infty\|\nabla\xi\|_\infty) \le CJ^3 2^{-J(\alpha-3/2)}$. Here each cross term has exponent $\le -(\alpha - 3/2)$ because $\alpha \ge 1$.

By Lemma 6.7(b) and Lemma 6.9(d), with $\|D\operatorname{div} V_J\|_\infty \le d\|D^2 V_J\|_\infty \le CJ2^{J(2-\alpha)_+}$ from Lemma 6.4(3), we get $\|Z\|_\infty \le CJ^2 2^{-J(\alpha-1/2)}$ and $\|\nabla Z\|_\infty \le C(\|\nabla G_\xi\|_\infty + \|G_\xi\|_\infty J2^{J(2-\alpha)_+}) \le CJ^3 2^{-J(\alpha-3/2)}$. Hence

$$\|\operatorname{div}(Z\nabla\xi)\|_\infty \le \|\nabla Z\|_\infty\|\nabla\xi\|_\infty + \|Z\|_\infty\|\Delta\xi\|_\infty \le CJ^4 2^{-J(2\alpha-1)} + CJ^3 2^{-J(2\alpha-1)} \le CJ^4 2^{-J(2\alpha-1)}.$$

Lemma 6.7(b) for $X = \hat V$ concludes the proof. □

Lemma 6.14. *$P(\mathcal{A}^c) \le Ce^{-cJ^2}$, uniformly in f.*

Proof. Each component $\partial^\iota \xi(x) = n^{-1/2}\mathbb{W}(\partial_x^\iota k_J(x,\cdot))$, $|\iota| = m$, is centred Gaussian. By (K3) its standard deviation is $\le C\bar\sigma_m$, where $\bar\sigma_m := n^{-1/2}2^{J(d/2+m)}$, and its canonical metric is $\le C\bar\sigma_m 2^J|x-x'|$. The covering numbers of D in this metric are therefore $\le (C\bar\sigma_m 2^J/\eta)^d$, and Dudley's bound gives $\mathbb{E}\sup_D|\partial^\iota\xi| \le C\bar\sigma_m\sqrt{J} \le J\bar\sigma_m/2$ for large J. The Borell–TIS inequality at deviation $J\bar\sigma_m/2$ gives $P(\sup_D|\partial^\iota\xi| > J\bar\sigma_m) \le 2e^{-cJ^2}$. A union bound over the finitely many ι with $1 \le |\iota| \le 3$ concludes the proof. □

6.9 Assembly

Proof of Theorem 2.1(ii). Let $r \le r_0$ (the minimum of the values required in Lemmas 6.1 and 6.4) and $J \ge J_0$. On $\mathcal{A}$ we have $\mathcal{A} \subset \mathcal{B}$, so (8) applies. Clipping only decreases $|\tilde f - f|$, so Proposition 6.11 and Lemma 6.13 give

$$\|\tilde f - f\|_2^2 \le 3|D|\big(CJ^2 2^{-J\alpha}\big)^2 + 3\|Z\|_2^2 + 3|D|\big(CJ^4 2^{-J(2\alpha-1)}\big)^2.$$

Off $\mathcal{A}$, $|\tilde f - f| \le 2r$ pointwise, so $\|\tilde f - f\|_2^2 \le 4r^2|D|$. Taking expectations, using Proposition 6.12, Lemma 6.14 and (4),

$$\mathbb{E}_f\|\tilde f - f\|_2^2 \le CJ^4 2^{-2J\alpha} + C2^{-2J\alpha} + CJ^8 2^{-2J(2\alpha-1)} + Ce^{-cJ^2} \le CJ^4 2^{-2J\alpha} \le C(\log n)^4 n^{-2\alpha/(2\alpha+d+3)},$$

uniformly over $\mathcal{F}(\alpha, r)$. The finitely many n with $J < J_0$ are covered by the trivial bound $4r^2|D|$. For non-integer α, Lemma 6.3 and (C3) produce no logarithm. Then the bias is $\le C2^{-J\alpha}$, and the remaining powers of J multiply terms of strictly smaller order. So the bound is $Cn^{-2\alpha/(2\alpha+d+3)}$. □

Theorem 2.1 follows from Sections 5 and 6.

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